06 — Testing Results and Test Cases

Item	Details
Course	DBMS Project — Semester 5
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Stack	React + Vite; Node.js + Express; MySQL; MongoDB
Repository	https://github.com/snehajisho/Satellite_Data_Provenance_Earth_Observatory-DBMS_Project

13. Testing Results

Testing is grouped into MySQL, MongoDB, API and frontend checks. SQL and MongoDB test scripts are included in the repository.

13.1 MySQL test cases

ID	Test case	Expected result / verification
SQL-01	Geographic location count	GEO_LOCATION contains 10 rows; count query returns PASS when equal to 10.
SQL-02	Observation count	OBSERVATION contains 20 rows.
SQL-03	Data product count	DATA_PRODUCT contains 20 rows.
SQL-04	Processing activity count	PROCESSING_ACTIVITY contains 10 rows.
SQL-05	Expected location coverage	All ten expected area names are present.
SQL-06	Observation-location integrity	No observation references a missing GEO_LOCATION row.
SQL-07	Product-observation integrity	No data product references a missing observation.
SQL-08	Product-location consistency	Product location matches the location of its linked observation.
SQL-09	Processing product references	Every activity references existing input and output products.
SQL-10	New product processing history	Five activities output products 12, 14, 16, 18 and 20.
SQL-11	Observation/location display	Inspection query displays observations with area and country.

ID	Test case	Expected result / verification
SQL-12	Product processing history display	Inspection query displays products with matching processing activity.
SQL-13	Cloud-cover range	No observation has cloud cover outside 0–100.

13.2 MongoDB test cases

ID	Test case	Expected result
TC-M01	Collection exists	observation_metadata exists.
TC-M02	Document count	Exactly 20 documents.
TC-M03	Observation IDs	IDs 1–20 are all present as 20 distinct IDs.
TC-M04	Duplicate IDs	No duplicate observation_id values.
TC-M05	Missing IDs	Every document has a non-null observation_id.
TC-M06	Location names	Every document has a non-empty location.area_name.
TC-M07	New location coverage	Mumbai, Delhi, Chennai, Helsinki and Moscow each have two documents.
TC-M08	Blank location names	No location.area_name is empty or null.
TC-M09	Image metadata coverage	Informational count of documents containing image_metadata.
TC-M10	Location mappings	Informational listing of observation IDs and the five new locations.

The script's PASS/FAIL counter evaluates TC-M01 through TC-M08. TC-M09 and TC-M10 print informational output. The team reported that all eight MongoDB assertions passed.

13.3 API test cases

ID	Test case	Expected result
API-01	Health endpoint	GET /health returns HTTP 200 and success=true.
API-02	Valid login	Valid credentials return HTTP 200, JWT token and user details; skipped if test credentials are absent.
API-03	Invalid login	HTTP 401.
API-04	Missing login fields	HTTP 400.
API-05	Unauthenticated observations	GET /observations returns HTTP 401 without a token.
API-06	Retrieve observations	Authenticated request returns 20 records.
API-07	Retrieve satellites	Authenticated request returns a non-empty array.

ID	Test case	Expected result
API-08	Observation by valid ID	GET /observations/1 returns HTTP 200 when authenticated.
API-09	Invalid observation ID	GET /observations/abc returns HTTP 400 when authenticated.
API-10	All metadata	GET /metadata returns 20 documents.
API-11	Metadata by ID	GET /metadata/1 returns HTTP 200.
API-12	Location search	Mumbai metadata search returns HTTP 200 and at least one record.
API-13	Missing location parameter	Location search without a location returns HTTP 400.
API-14	Invalid metadata ID	GET /metadata/abc returns HTTP 400.
API-15	Unauthenticated provenance	GET /provenance/1 returns HTTP 401 without a token.
API-16	Authenticated provenance	Returns HTTP 200 and a data array.
API-17	Unknown endpoint	Unknown route returns HTTP 404.

The API test file uses `node:test` and `assert/strict`. It reads `API_BASE_URL` (default `http://localhost:3001/api`), `API_TEST_EMAIL` and `API_TEST_PASSWORD` from environment variables. Authenticated tests are skipped if credentials are unavailable or login does not yield a token. The team reported 17/17 API tests passing with the test environment configured.

13.4 Frontend manual test cases

ID	Manual scenario	Steps and expected behaviour
FE-01	Home page	Open Vite URL; header, hero, search and navigation render.
FE-02	Location search	Search Mumbai; quality view opens and matching metadata is displayed.
FE-03	Location suggestions	Choose a suggested location; search is submitted and results update.
FE-04	Registration validation	Submit missing fields and a password shorter than eight characters; validation appears.
FE-05	Registration success	Register with valid username, email and password; success message appears.
FE-06	Login	Sign in with a valid account; authenticated navigation and role appear.

ID	Manual scenario	Steps and expected behaviour
FE-07	Logout	Sign out; local session is cleared and protected pages are no longer accessible.
FE-08	Expired session	Use an invalid/expired token; UI returns to auth state and shows an error.
FE-09	Satellite explorer	Open Satellites, search/select a satellite and inspect details.
FE-10	Observation explorer	Open Observations and inspect list and selected record details.
FE-11	Analytics	Open Analytics and verify displayed summaries from loaded data.
FE-12	Provenance	Open Provenance while authenticated and inspect processing history.
FE-13	Quality view	Open Quality and inspect location metadata and quality indicators.
FE-14	Image loading	Verify previews resolve through the backend /images route.
FE-15	Role restrictions	Test USER and ADMIN accounts; confirm server authorization behaviour.